CS 577 — Deep Learning — Homework 4

Read these instructions carefully:

- In the LaTeX source code, type your answer in between "`%%% BEGIN ANSWER`" and "`%%% END ANSWER`". For advanced LaTeXusers, you can use your custom macros if you wish by placing them between "`%%% BEGIN MACROS`" and "`%%% END MACROS`" in the header. Do not modify anything else.
- Turn in both your `.tex` file and the generated `.pdf` file.

1 Backpropagation

[5] point(s) — part a:
Answer the question in Section 1.3.7 of `backpropagation.pdf`: How can we calculate $\frac{\partial f}{\partial z_4}(x, y)$ given correct value of $\frac{\partial f}{\partial z_5}(x, y)$?

Answer:
On by applying chain rule on $\frac{\partial f}{\partial z_4}$, we get ,

$$\frac{\partial f}{\partial z_4} = \frac{\partial f}{\partial z_5} \cdot \frac{\partial z_5}{\partial z_4}$$

Given,

$$z_5 = \exp(z_4)$$

The partial derivative of z_5 with respect to z_4 is :

$$\frac{\partial z_5}{\partial z_4} = \frac{\partial}{\partial z_4}(\exp(z_4)) = \exp(z_4) = z_5$$

$\frac{\partial z_5}{\partial z_4} = z_5$, we substituting this into the chain rule of $\frac{\partial f}{\partial z_4} expression$:

$$\frac{\partial f}{\partial z_4} = \frac{\partial f}{\partial z_5} \cdot z_5$$

The final expression for $\frac{\partial f}{\partial z_4}$ is:

$$\boxed{\frac{\partial f}{\partial z_4} = \frac{\partial f}{\partial z_5} \cdot z_5}$$

[5] point(s) — part b:
Answer the question in Section 1.3.8 of `backpropagation.pdf`: What is currently stored in `z3.grad` right before `z4._backward()` is called?

Answer:
`z3.grad` is calculated as,

$$\frac{\partial f}{\partial z_3}(x,y) = \frac{\partial f}{\partial z_8}(x,y)\frac{\partial z_8}{\partial z_3} = \frac{\partial f}{\partial z_8}(x,y)\frac{\partial (z_7 * z_3)}{\partial z_3} = \underbrace{z_7 \frac{\partial f}{\partial z_8}(x,y)}_{\text{term 1}} + \underbrace{\frac{\partial f}{\partial z_8}(x,y)\frac{\partial z_7}{\partial z_3}}_{\text{term 2}}$$

Before `z4.backward()` is called:

$$z8 = z7 \times z3$$

the gradient for both $z7$ and $z3$. When backpropagating through $z8$, the gradient of the loss (or output) with respect to $z3$ is calculated using the product rule:

$$\frac{\partial f}{\partial z3}(x,y) = \frac{\partial f}{\partial z8}(x,y) \times \frac{\partial z8}{\partial z3} = \frac{\partial f}{\partial z8}(x,y) \times z7$$

Before `z4.backward()` is called, only term 1 is calculated.

Right before `z4.backward()` is called, $z3.grad$ has the first term, $z7 \times \frac{\partial f}{\partial z8}$. The second term, which depends on the relationship between $z3$ and $z7$, will only be finalized once backpropagation reaches $z4$.

Final value of $z3.grad$:

Initially, the gradient in $z3.grad$ is:

$$z3.grad = z7 \times \frac{\partial f}{\partial z8}(x,y)$$

Since $z8$ is equal to $f(x,y)$ (the output), and the gradient of the loss with respect to $z8$ is initialized to 1. If we change z8 = f(x,y) by a tiny amount, then f(x,y) will change by that same amount (tautology). Thus,:

$$\frac{\partial f}{\partial z8}(x,y) = 1$$

So, before `z4.backward()` is called, the value of $z3.grad$ is:

$$z3.grad = z7$$

2 Gradient descent with `ag.Scalar`

[10] point(s) — part a: This is a programming exercise. See `hw4.ipynb`

3 Transformer with `ag.Scalar`

[Bonus 20] point(s) — part a: This is a programming exercise. See `hw4.ipynb`